

Soret and Dufour Effects on MHD Free Convection Flow of a Chemically Reacting Fluid Past over a Stretching Sheet with Heat Source/Sink

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Abstract

A mathematical model is presented for a two-dimensional, steady, viscous, incompressible, electrically conducting and laminar MHD free convection flow with Soret and Dufour effects in the presence of porous medium and heat generation/absorption. The governing differential equations of the problem have been transformed into a system of non-dimensional differential equations, which are then solved numerically using a fourth-order Runge-Kutta method along with shooting technique. The velocity and temperature distributions are discussed numerically and presented through graphs. The numerical values of skin-friction coefficient and Nusselt number at the plate are derived, discussed numerically for various values of physical parameters and presented through Tables. As the heat flux exponent parameter or suction/injection parameter increases, both the local skin-friction coefficient and Sherwood number increase, whereas the Nusselt number decreases. It is observed that the local skin-friction coefficient and local Nusselt number decrease, whereas Sherwood number increases.

Keywords

MHD, Free Convection, Heat and Mass Transfer, Soret and Dufour Effects, Porous Medium, Heat Generation/Absorption

1. Introduction

The study of Magneto hydrodynamic (MHD) flows have stimulated extensive attention due to its significant applications in three different subject areas, such as astrophysical, geophysical and engineering problems. Free convection in electrically conducting fluids through an external magnetic field has been a subject of considerable research interest of a large number of scholars for a long time due to its diverse applications in the fields such as nuclear reactors, geothermal engineering, liquid metals and plasma flows, among others. Fluid flow control under magnetic forces is also applicable in magneto hydrodynamic generators and a host of magnetic devices used in industries. Steady and transient free convection of an electrically conducting fluid from a vertical plate in the presence of magnetic field was studied by Gupta [1]. Lykoudis [2] investigated natural convection of an electrically conducting fluid with a magnetic field. Palani and Srikanth [3] studied the MHD flow of an electrically conducting fluid over a semi-infinite vertical

plate under the influence of the transversely applied magnetic field. Makinde [4] investigated the MHD boundary layer flow with the heat and mass transfer over a moving vertical plate in the presence of magnetic field and convective heat exchange at the surface. Takhar et al. [5] computed flow and mass transfer on a stretching sheet under the consideration of magnetic field and chemically reactive species. They focused that the energy flux can be produced by both the temperature gradient and concentration gradient.

The energy flux caused by concentration gradient is called Dufour effect and the same by temperature gradient is called the Soret effect. These effects have a vital role in the high temperature and high concentration gradient. The significant Soret effect in convective transport in clear fluids has been found in the work of Bergaman and Srinivasan [6] and Zimmerman et al. [7]. The effect of magnetic field on heat and mass transfer from vertical surfaces in porous media considering Soret and Dufour effects have been performed by Postelnicu [8]. Ahammad [9] discussed free convective heat and mass transfer of an incompressible, electrically conducting fluid over a stretching sheet in the presence of

suction and injection with thermal-diffusion and diffusion-thermo effects. A study has been carried out to analyze the combined effects of Soret and Dufour on unsteady MHD non-Darcy mixed convection over a stretching sheet embedded in a saturated porous medium in the presence of thermal radiation, viscous dissipation and first-order chemical reaction by Pal and Mondal [10]. Alan and Rahman [11], examined Dufour and Soret effects on mixed convection flow past a vertical porous flat plate with variable suction embedded in a porous medium for a hydrogen-air mixture as the nonchemical reacting fluid pair. Gaikwad et al. [12], investigated the onset of double diffusive convection in a two component couple stress fluid layer with Soret and Dufour effects using both linear and non-linear stability analysis.

Steady heat transfer analysis in the presence of heat source has been receiving wide attention among the researchers due to its applications in polymer extrusion process, metallurgical process, drawing of artificial fibers etc. Xu [13] studied the free convective heat transfer characteristics in an electrically conducting fluid near an isothermal sheet with internal heat generation or absorption. Abel and Mahesha [14] have investigated the effects of thermal conductivity, non-uniform heat source and viscous dissipation in the presence of thermal radiation on the flow and heat transfer in viscoelastic fluid over a stretching sheet, which is subjected to an external magnetic field. Pillai et al. [15] investigated the effects of work done by deformation in viscoelastic fluid in porous media with uniform heat source, Hayat et al. [16] also investigated the effects of work done by deformation in second grade fluid with partial slip condition, in this no account of heat source has been taken into consideration and Khan et al. [17] also investigated the effect of work done by deformation in Walter's liquid B but with uniform heat source. Moreover, Radiation and mass transfer effects on MHD free convection fluid flow embedded in a porous medium with heat generation/absorption was studied by Shankar et al. [18]. Suneetha et al. [19] studied Radiation and Mass Transfer Effects on MHD Free Convective Dissipative Fluid in the Presence of Heat Source/Sink,

Combined heat and mass transfer problems with chemical reaction are of importance in many processes and have, therefore, received a considerable amount of attention in recent years. In processes such as drying, evaporation at the surface of a water body, energy transfer in a wet cooling tower and the flow in a desert cooler, heat and mass transfer occur simultaneously. Possible applications of this type of flow can be found in many industries. For example, in the power industry, among the methods of generating electric power is one in which electrical energy is extracted directly from a moving conducting fluid. Many practical diffusive operations involve the molecular diffusion of a species in the presence of chemical reaction within or at the boundary. The study of heat and mass transfer with chemical reaction is of great practical importance to engineers and scientists because of its almost universal occurrence in many branches of science and engineering. Al-Azab [20] numerically studied the transient MHD free convective heat and mass transfer over a moving

vertical surface in the presence of a homogeneous chemical reaction of first order. Chambre and Young [21] have presented a first order chemical reaction in the neighborhood of a horizontal plate. Dekha et al. [22] investigated the effect of the first order homogeneous chemical reaction on the process of an unsteady flow past a vertical plate with a constant heat and mass transfer. Muthukumaraswamy and Ganesan [23] studied effect of the chemical reaction and injection on the flow characteristics in an unsteady upward motion of an isothermal plate. Suneetha and Bhaskar Reddy [24] studied Radiation and Darcy effects on unsteady MHD heat and mass transfer flow of a chemically reacting fluid past an impulsively started vertical plate with heat generation.

Chamkha [25] studied the MHD flow of a numerical of uniform stretched vertical permeable surface in the presence of heat generation/absorption and a chemical reaction. Seddeek et al. [26] analyzed the effects of chemical reaction, radiation and variable viscosity on hydro magnetic mixed convection heat and mass transfer Hiemenz flow through porous media. Kandasamy et al. [27] studied Chemical Reaction, Heat and Mass Transfer on MHD Flow over a Vertical Stretching Surface with Heat Source and Thermal Stratification Effects.

However, only a limited attention has been paid to the study of effect of chemical reaction over stretching surfaces which is useful in many industrial applications processes such as manufacturing of ceramics, food processing and polymer production, drying, evaporation, energy transfer in a cooling tower and the flow in a desert cooler, heat and mass transfer occur simultaneously. Kameswaran et al. [28] investigated the convective heat and mass transfer in nano fluid flow over a stretching sheet when subjected to hydro magnetic, viscous dissipation, chemical reaction and Soret effects. The object of the present chapter is to analyze the two-dimensional, steady, viscous, incompressible, electrically conducting and laminar MHD free convection flow with Soret and Dufour effects by taking heat generation/absorption and chemical reaction into account. Liu [29] studied heat and mass transfer in a MHD flow past a stretching sheet including the chemically reactive species of order one and internal heat generation or absorption. The governing boundary layer equations have been transformed to a two-point boundary value problem in similarity variables and the resultant problem is solved numerically using the Runge-Kutta method with shooting technique. The effects of various governing parameters on the fluid velocity, temperature, concentration, skin-friction coefficient, Nusselt number and Sherwood number are shown in figures and tables and analyzed in detail.

2. Mathematical Analysis

A steady, two-dimensional, laminar MHD free convective, viscous, chemically reacting and incompressible along a linearly stretching semi-infinite sheet in the presence of heat generation/absorption is considered. In the present problem, it can be considered that the flow is steady, two-dimensional, laminar MHD free convective, viscous and incompressible

along a linearly stretching semi-infinite sheet. The surface is supposed to be permeable which moves with velocity $u_w(x) = bx$ and $v_w(x)$ represents the permeability of the porous surface. Fluid suction is imposed at the stretching surface. The x -axis runs along the stretching surface in the direction of motion with the slot as the origin and the y -axis is measured normally from the sheet to the fluid. The applied magnetic field is primarily in the y -direction and varies in strength as a function of x and is defined as $B = (0, B(x))$. Moreover, the electrical conductivity σ is assumed to have the form as $\sigma = \sigma_0 u$, where σ_0 is a constant.

According to the usual Boussinesq and boundary-layer approximation, the governing equations for this problem can be written as follows:

Continuity equation

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \quad (1)$$

Momentum equation

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = \nu \frac{\partial^2 u}{\partial y^2} - \frac{\sigma(B(x))^2}{\rho} u + g\beta_T(T - T_\infty) + g\beta_c(C - C_\infty) \quad (2)$$

Energy equation

$$u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} = \frac{k}{\rho c_p} \frac{\partial^2 T}{\partial y^2} + \frac{D_m k_T}{c_s c_p} \frac{\partial^2 C}{\partial y^2} + Q_0(T - T_\infty) \quad (3)$$

Species equation

$$u \frac{\partial C}{\partial x} + v \frac{\partial C}{\partial y} = D_m \frac{\partial^2 C}{\partial y^2} + \frac{D_m k_T}{T_m} \frac{\partial^2 T}{\partial y^2} - k_0(C - C_\infty) \quad (4)$$

It is appropriate to suppose that the applied magnetic field strength $B(x)$ has the form (Helmy [30]):

$$B(x) = \frac{B_0}{\sqrt{x}}, \quad B_0 \text{ is constant.} \quad (5)$$

Using equation (5), the second term in equation (2) can be rewritten as:

$$\frac{\sigma(B(x))^2 u}{\rho} = \frac{\sigma_0 B_0^2 u^2}{\rho x}, \quad \text{where } \sigma = \sigma_0 u. \quad (6)$$

Therefore by means of equation (6), equation (2) reduces to

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = \nu \frac{\partial^2 u}{\partial y^2} - \frac{\sigma_0 B_0^2}{\rho x} u^2 + g\beta_T(T - T_\infty) + g\beta_c(C - C_\infty) - \frac{\nu}{k_0} u \quad (7)$$

The boundary conditions for the velocity, temperature and concentration fields are

$$u = u_w(x) = bx, v = \pm v_w(x), -k \frac{\partial T}{\partial y} = q_w = A_1 x^r,$$

$$-D_m \frac{\partial C}{\partial y} = M_w = A_2 x^r$$

at $y = 0$

$$u \rightarrow 0, T \rightarrow T_\infty, C \rightarrow C_\infty \text{ as } y \rightarrow \infty \quad (8)$$

where u , v , T and C are the fluid x -component of velocity, y -component of velocity, temperature and concentration respectively, ν is the fluid kinematics viscosity, ρ - the density, σ - the electric conductivity of the fluid, r - the heat flux exponent parameter, β_T and β_c - the coefficients of thermal and concentration expansions respectively, k - the thermal conductivity, k_0 - the chemical reaction rate constant, Q_0 - the heat generation/absorption, C_∞ - the free stream concentration, B_0 - the magnetic induction, D_m - the mass diffusivity and g is the gravitational acceleration.

The mass concentration equation (1) is satisfied by the Cauchy-Riemann equations

$$u = \frac{\partial \psi}{\partial y}, v = -\frac{\partial \psi}{\partial x}. \quad (9)$$

where $\psi(x, y)$ is the stream function.

To transform equations (7), (3) and (4) into a set of ordinary differential equations, the following similarity transformations and dimensionless variables are introduced.

$$\eta = y \sqrt{\frac{b}{\nu}}, \quad \psi = x \sqrt{\nu b} f(\eta), \quad T - T_\infty = \frac{A_1 x^r}{k} \left(\frac{\nu}{b} \right)^{\frac{1}{2}} \theta(\eta),$$

$$C - C_\infty = \frac{A_1 x^r}{k} \left(\frac{\nu}{b} \right)^{\frac{1}{2}} \phi(\eta), \quad M = \frac{\sigma B_0^2 x}{\rho u_w(x)}, \quad Gr = \frac{g \beta_T q_w x^4}{k \nu^2 \text{Re}_x^{5/2}},$$

$$Gc = \frac{g \beta_c M_w x^4}{D_m \nu^2 \text{Re}_x^{5/2}}, \quad Du = \frac{D_m k_T M_w}{\nu c_s c_p q_w}, \quad Sr = \frac{D_m k_T q_w}{\nu T_m M_w},$$

$$Q = \frac{Q_0}{\rho c_p b}, \quad kr = \frac{k_0}{b},$$

$$\text{Pr} = \frac{\nu \rho c_p}{k}, \quad \text{Sc} = \frac{\nu}{D_m}, \quad \text{Re}_x = \frac{u_w(x)x}{\nu}. \quad (10)$$

where η - similarity variable, f - dimensionless stream function, θ - dimensionless temperature, ϕ - dimensionless concentration, M - the Magnetic field parameter, Gr - the thermal Grashof number, Gc - the solutal Grashof number, Du - the Dufour number, Sr - the Soret number, Q - the heat generation/absorption parameter, Pr - the Prandtl number, Sc - the Schmidt number, kr - chemical reaction parameter and Re_x - the local Reynolds number.

In view of equations (9) and (10), Equations (7), (3) and (4)

transform into

$$f''' + ff'' - f'^2 + Gr\theta + Gc\phi - Mf'^2 = 0 \quad (11)$$

$$\theta'' - rPr f'\theta + Pr f\theta' + Pr Du\phi'' + Q\theta = 0 \quad (12)$$

$$\phi'' - rScf'\phi + Scf\phi' + ScSr\theta'' - Sckr\phi = 0 \quad (13)$$

The corresponding boundary conditions are

$$f = fw, f' = 1, \theta' = -1, \phi' = -1 \text{ at } \eta = 0$$

$$f' = 0, \theta = \phi = 0 \text{ as } \eta \rightarrow \infty \quad (14)$$

where the prime symbol represents the derivative with respect to η and the dimensionless wall mass transfer coefficient defined as $fw = -v_w / (bv)^{1/2}$, whose positive and negative value indicates wall suction and wall injection respectively.

From the numerical computation, the local skin-friction coefficient, the local Nusselt number and Sherwood number which are, respectively, proportional to $f''(0)$, $\frac{1}{\theta(0)}$ and $\frac{1}{\phi(0)}$ are worked out and their numerical values are presented in a tabular form.

3. Solution of the Problem

The set of coupled non-linear governing boundary layer equations (11) - (13) together with the boundary conditions (14) are solved numerically by using Runge-Kutta fourth order technique along with shooting method. First of all, higher order non-linear differential Equations (11) - (13) are converted into simultaneous linear differential equations of first order and they are further transformed into initial value problem by applying the shooting technique (Jain *et al* [31]). The resultant initial value problem is solved by employing Runge-Kutta fourth order technique. The step size $\Delta\eta = 0.05$ is used to obtain the numerical solution. From the process of numerical computation, the skin-friction coefficient, the Nusselt number and the Sherwood number, which are respectively proportional to $f''(0)$, $\frac{1}{\theta(0)}$ and $\frac{1}{\phi(0)}$, are also sorted out and their numerical values are presented in a tabular form.

4. Results and Discussion

The governing equations (11) - (13) subject to the boundary conditions (14) are integrated as described in section 3. Numerical results are reported in the Tables 1. The Prandtl number is taken to be $Pr = 0.71$ which corresponds to air, the value of Schmidt number (Sc) were chosen to be $Sc = 0.24, 0.62, 0.78, 2.62$, representing diffusing chemical species of most common interest in air like H_2 , H_2O , NH_3 and Propyl Benzene respectively.

The effects of various parameters on velocity profiles in the boundary layer are depicted in Figs. 1-10. In Fig. 1 the

effect of increasing the magnetic field strength on the momentum boundary layer thickness is illustrated. It is now a well-established fact that the magnetic field presents a damping effect on the velocity field by creating drag force that opposes the fluid motion, causing the velocity to decrease. Similar trend of slight decrease in the fluid velocity near the vertical plate is observed with an increase in suction/injection parameter (fw) (see in Fig. 2). Fig. 3 illustrates the effect of the thermal Grashof number (Gr) on the velocity field. The thermal Grashof number signifies the relative effect of the thermal buoyancy force to the viscous hydrodynamic force. The flow is accelerated due to the enhancement in buoyancy force corresponding to an increase in the thermal Grashof number, i.e. free convection effects. It is noticed that the thermal Grashof number (Gr) influence the velocity field almost in the boundary layer when compared to far away from the plate. It is seen that as the thermal Grashof number (Gr) increases, the velocity field increases. The effect of mass (solutal) Grashof number (Gc) on the velocity is illustrated in Fig. 4. The mass (solutal) Grashof number (Gc) defines the ratio of the species buoyancy force to the viscous hydrodynamic force. It is noticed that the velocity increases with increasing values of the solutal Grashof number.

Fig. 5 illustrates the effect of the Schmidt number (Sc) on the velocity. The Schmidt number (Sc) embodies the ratio of the momentum diffusivity to the mass (species) diffusivity. It physically relates the relative thickness of the hydrodynamic boundary layer and mass-transfer (concentration) boundary layer. It is noticed that as Schmidt number (Sc) increases the velocity field decreases. Similar trend of slight decrease in the fluid velocity near the vertical plate is observed with an increase in Prandtl number (Pr) (see in Fig. 6). Fig. 7 shows the variation of the velocity boundary-layer with the heat generation/absorption parameter (Q). It is noticed that the velocity boundary layer thickness increases with an increase in the generation/absorption parameter. Fig. 8 shows the variation of the velocity boundary-layer with the Dufour number (Du). It is observed that the velocity boundary layer thickness increases with an increase in the Dufour number. Fig. 9 shows the variation of the velocity boundary-layer with the Soret number (Sr). It is found that the velocity boundary layer thickness increases with an increase in the Soret number. Fig. 10 shows the variation of the velocity boundary-layer with the chemical reaction parameter (kr). It is found that the velocity boundary layer thickness decreases with an increase in the chemical reaction parameter.

The effects of various parameters on temperature profiles in the boundary layer are depicted in Figs. 11-20. The effect of the magnetic parameter (M) on the temperature is illustrated in Fig. 11. It is observed that as the magnetic parameter increases, the temperature increases. Fig. 12 illustrates the effect of the suction/injection parameter (fw) on the temperature. It is noticed that as suction parameter increases, the temperature decreases. From Figs. 13 and 14, it is observed that the thermal boundary layer thickness decreases with an increase in the thermal or Solutal Grashof

numbers (Gr or Gc). Fig. 15 illustrates the effect of Schmidt number (Sc) on the temperature. It is noticed that as the Schmidt number (Sc) increases an increasing trend in the temperature field occurs. Much of significant contribution of Schmidt number (Sc) is noticed as we move far away from the plate.

Fig. 16 illustrates the effect of the Prandtl number (Pr) on the temperature. It is noticed that as the Prandtl number increases, the temperature decreases. Fig. 17 illustrates the effect of the heat generation/absorption parameter (Q) on the temperature. It is noticed that as the heat generation/absorption parameter increases, the temperature increases. Fig. 18 shows the variation of the thermal boundary-layer with the Dufour number (Du). It is noticed that the thermal boundary layer thickness increases with an increase in the Dufour number. Fig. 19 shows the variation of the thermal boundary-layer with the Soret number (Sr). It is observed that the thermal boundary layer thickness decreases with an increase in the Soret number. The effect of chemical reaction parameter (kr) on the temperature field is illustrated in Fig. 20. As the chemical reaction parameter increases the thermal boundary layer is found to be increasing.

Figs. 21 - 27 depict chemical species concentration against span wise coordinate η for varying values of physical parameters in the boundary layer. The effect of suction/injection parameter (fw) on the concentration field is illustrated Fig. 21. As the suction/injection parameter increases the concentration is found to be decreasing. However, as we move away from the boundary layer, the effect is not significant. The effect of buoyancy parameters (Gr , Gc) on the concentration field is illustrated in Figs. 22 and 23. It is noticed that the concentration boundary layer thickness decreases with an increase in the thermal or Solutal Grashof numbers (Gr or Gc). Fig. 24 illustrates the effect of Schmidt number (Sc) on the concentration. It is noticed that as the Schmidt number increases, there is a decreasing trend in the concentration field.

Fig. 25 shows the variation of the concentration boundary-layer with the Dufour number (Du). It is observed that the concentration boundary layer thickness decreases with an increase in the Dufour number. Fig. 26 shows the variation of the concentration boundary-layer with the Soret number (Sr). It is found that the concentration boundary layer thickness increases with an increase in the Soret number. Fig. 27 shows the variation of the thermal boundary-layer with the chemical reaction parameter (kr). It is observed that the thermal boundary layer thickness decreases with an increase in the chemical reaction parameter.

From Table 1, it is observed that the local skin-friction coefficient, local heat and mass transfer rates at the stretching sheet increases with an increase in the buoyancy forces. It is noticed that the local skin-friction coefficient, local heat and mass transfer rates at the sheet decreases with an increase in the Magnetic parameter. As the Dufour number or heat generation/absorption parameter increases, both the skin-friction and Sherwood number increase, whereas the Nusselt number decreases. It is observed that the Skin-friction

and local heat transfer rate at the sheet increase, whereas Sherwood number decreases with an increase in the Soret number. It is found that the local heat and mass transfer rate at the sheet increases, but Skin-friction coefficient decreases with an increase in the suction/injection parameter. It is observed that the Skin-friction and local heat transfer rate at the sheet increase, whereas Sherwood number decreases with an increase in the Schmidt number or chemical reaction parameter.

5. Conclusions

The present chapter analyzes the two-dimensional, steady, viscous, incompressible, electrically conducting and laminar MHD free convection flow with Soret and Dufour effects by taking heat generation/absorption and chemical reaction in to account. The governing equations are approximated to a system of non-linear ordinary differential equations by similarity transformation. Numerical calculations are carried out for various values of the dimensionless parameters of the problem. The results are presented graphically and the conclusion is drawn that the flow field and other quantities of physical interest are significantly influenced by these parameters. It is noticed that as the velocity and temperature increase by rising heat generation/absorption parameter. The velocity and concentration profiles decrease whereas temperature profiles increase by rising chemical reaction parameter. The results for the prescribed skin friction, Nusselt number and Sherwood number at the stretching sheet are presented and discussed. It is found that the local skin-friction coefficient, local heat and mass transfer rates at the sheet decrease with an increase in the Magnetic parameter. As the heat flux exponent parameter or suction/injection parameter increases, both the local Skin-friction coefficient and Sherwood number increase, whereas the Nusselt number decreases. It is observed that the local skin-friction coefficient and local Nusselt number decrease, whereas Sherwood number increases.

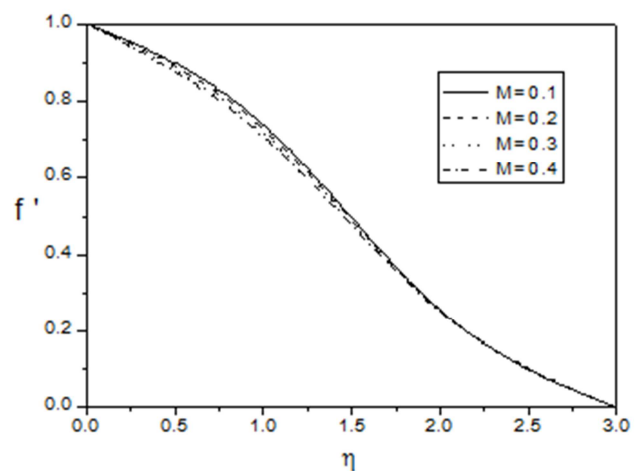


Fig. 1. Variation of the velocity f' with M for $Pr=0.71$, $Sc=0.22$, $Gr=Gc=1$, $Q=r=0.01$, $Sr=0.6$, $Du=0.2$, $kr=0.5$, $fw=0.1$.

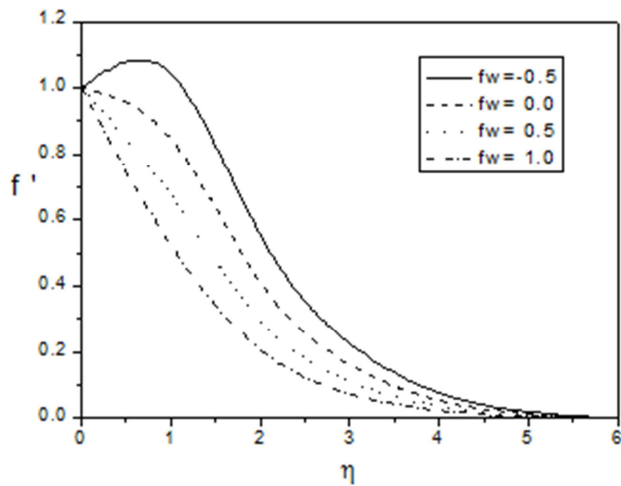


Fig. 2. Variation of the velocity f' with fw for $Pr=0.71$, $Sc=0.22$, $Gr=Gc=1$, $Q=r=0.01$, $Sr=0.6$, $Du=0.2$, $kr=0.5$, $M=0.1$.

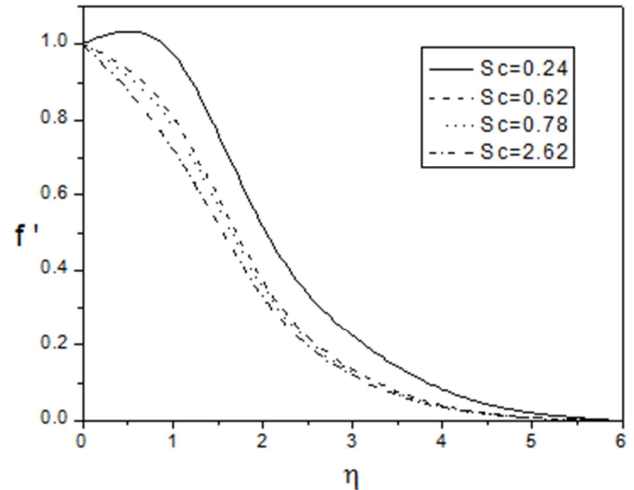


Fig. 5. Variation of the velocity f' with Sc for $Pr=0.71$, $Gr=Gc=1$, $Q=r=0.01$, $Sr=0.6$, $Du=0.2$, $kr=0.5$, $kr=fw=0.1$.

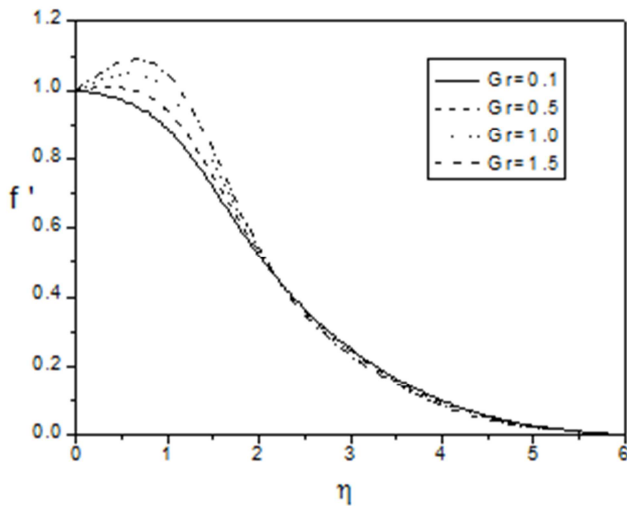


Fig. 3. Variation of the velocity f' with Gr for $Pr=0.71$, $Sc=0.22$, $Gc=1$, $Q=r=0.01$, $Sr=0.6$, $Du=0.2$, $kr=0.5$, $M=fw=0.1$.

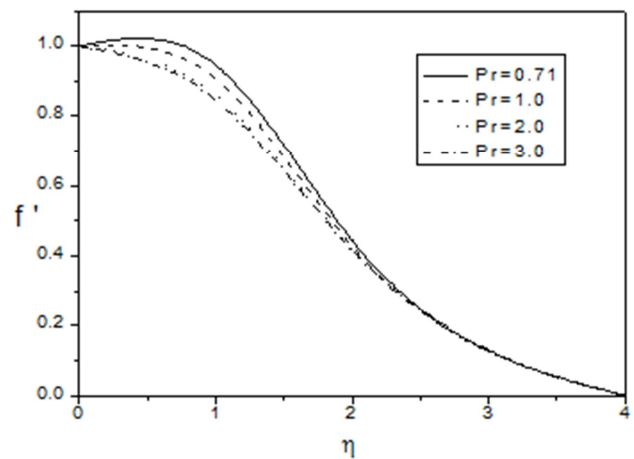


Fig. 6. Variation of the velocity f' with Pr for $Sc=0.22$, $Gr=Gc=1$, $Q=r=0.01$, $Sr=0.6$, $Du=0.2$, $kr=0.5$, $M=fw=0.1$.

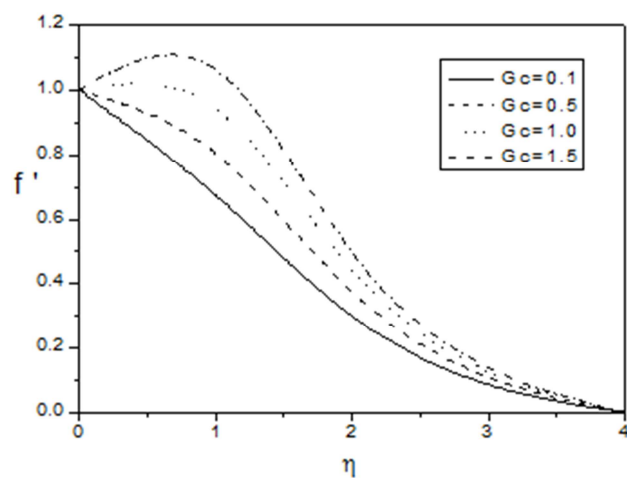


Fig. 4. Variation of the velocity f' with Gc for $Pr=0.71$, $Sc=0.22$, $Gr=1$, $Q=r=0.01$, $Sr=0.6$, $Du=0.2$, $kr=0.5$, $M=fw=0.11$.

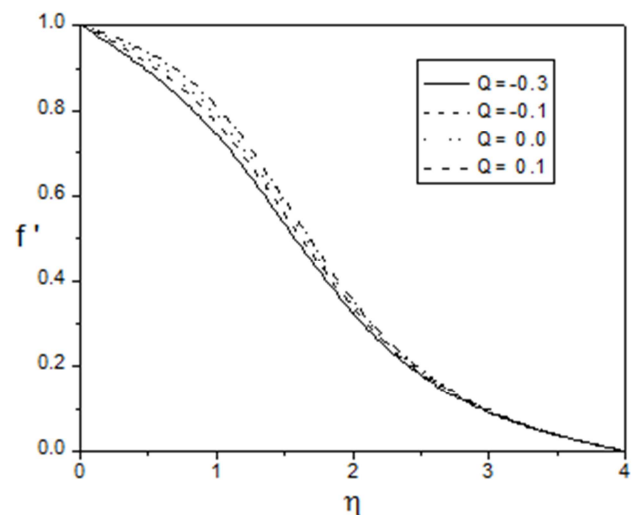


Fig. 7. Variation of the velocity f' with Q for $Pr=0.71$, $Sc=0.22$, $Gr=Gc=1$, $r=0.01$, $Sr=0.6$, $Du=0.2$, $kr=0.5$, $M=fw=0.1$.

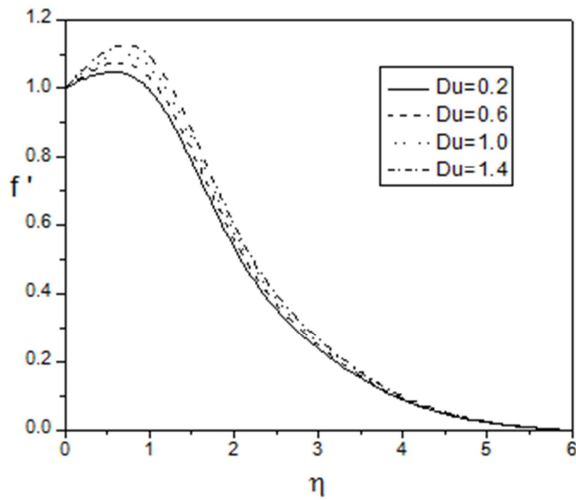


Fig. 8. Variation of the velocity f' with Du for $Pr=0.71$, $Sc=0.22$, $Gr=Gc=1$, $Q=r=0.01$, $Sr=0.6$, $kr=0.5$, $M=fw=0.1$.

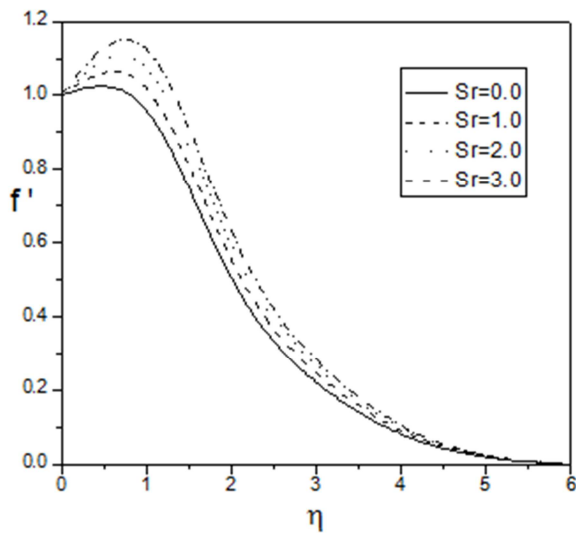


Fig. 9. Variation of the velocity f' with Sr for $Pr=0.71$, $Sc=0.22$, $Gr=Gc=1$, $Q=r=0.01$, $Du=0.2$, $kr=0.5$, $M=fw=0.1$.

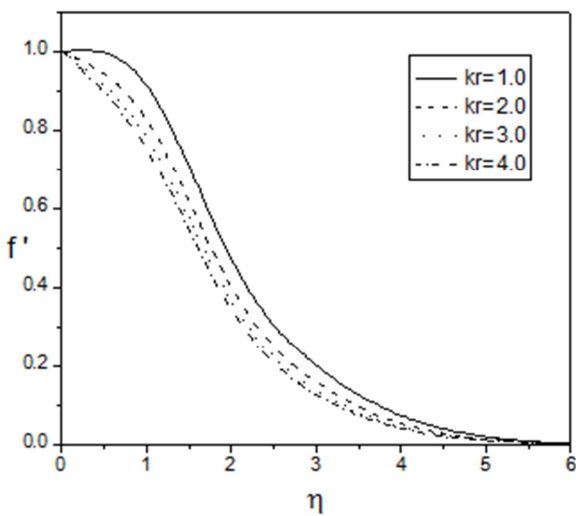


Fig. 10. Variation of the velocity f' with kr for $Pr=0.71$, $Sc=0.22$, $Gr=Gc=1$, $Q=r=0.01$, $Sr=0.6$, $Du=0.2$, $M=fw=0.1$.

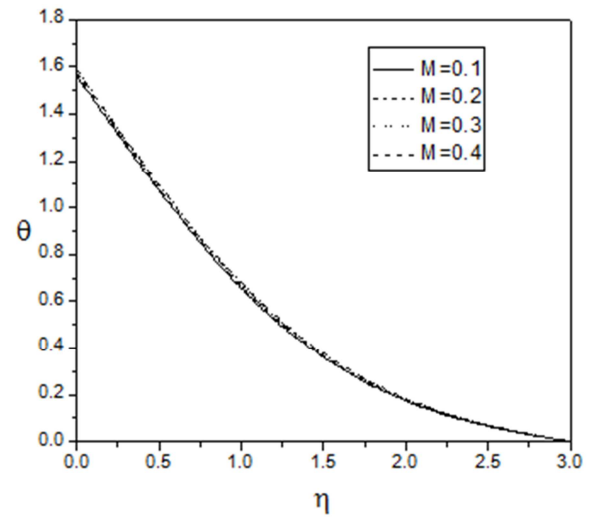


Fig. 11. Variation of the temperature θ with M for $Pr=0.71$, $Sc=0.22$, $Gr=Gc=1$, $Sr=0.6$, $Du=0.2$, $Q=r=0.01$, $kr=0.5$, $fw=0.1$.

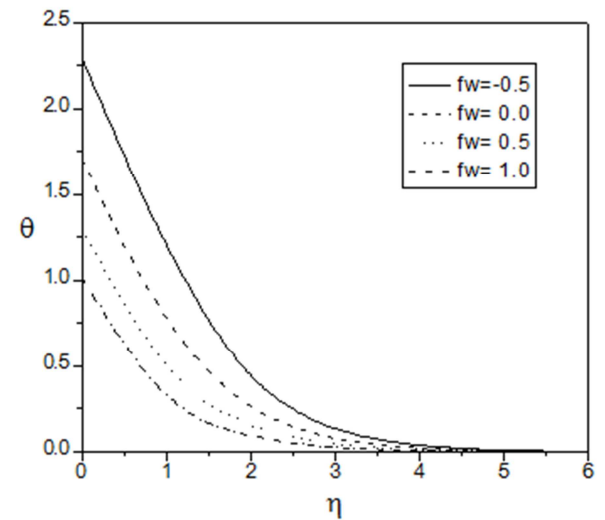


Fig. 12. Variation of the temperature θ with fw for $Pr=0.71$, $Sc=0.22$, $Gr=Gc=1$, $Sr=0.6$, $Du=0.2$, $Q=r=0.01$, $kr=0.5$, $M=0.1$.

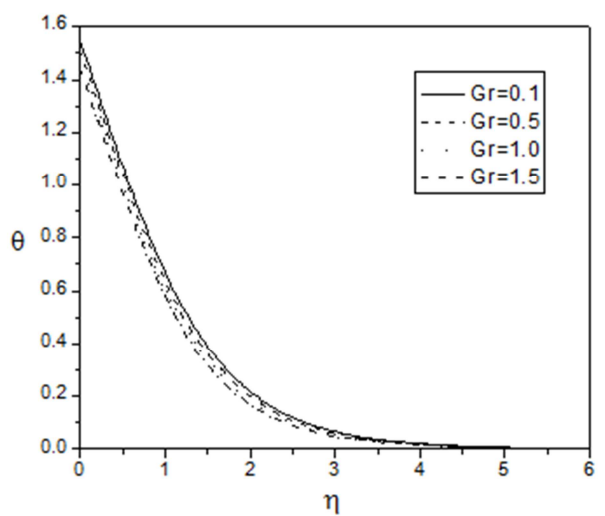


Fig. 13. Variation of the temperature θ with Gr for $Pr=0.71$, $Sc=0.22$, $Gc=1$, $Sr=0.6$, $Du=0.2$, $Q=r=0.01$, $kr=0.5$, $M=fw=0.1$.

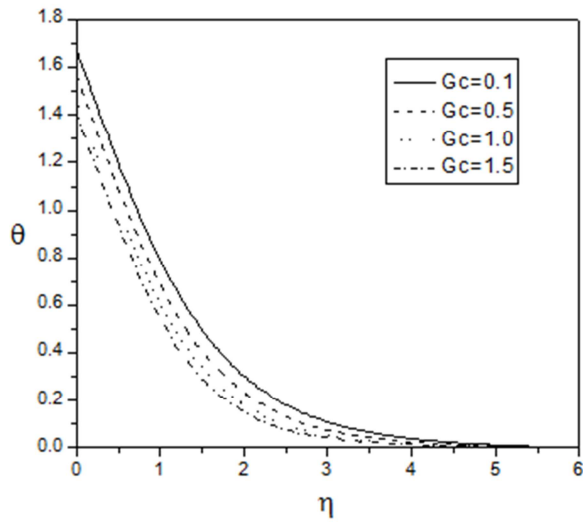


Fig. 14. Variation of the temperature θ with Gc for $Pr=0.71$, $Sc=0.22$, $Gr=1$, $Sr=0.6$, $Du=0.2$, $Q=r=0.01$, $kr=0.5$, $M=fw=0.1$.

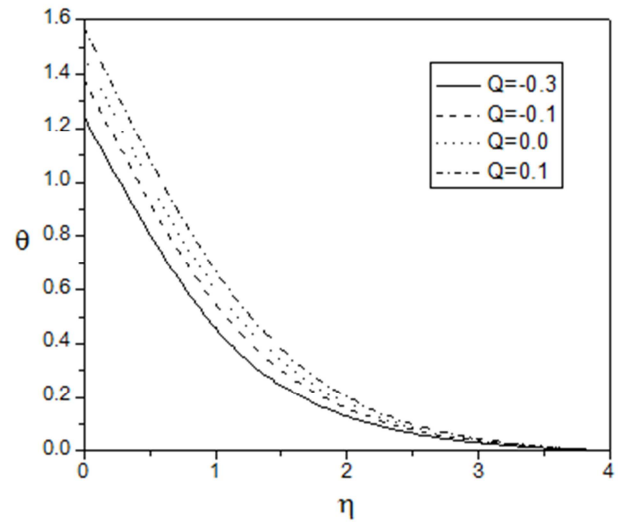


Fig. 17. Variation of the temperature θ with Q for $Pr=0.71$, $Sc=0.22$, $Gr=Gc=1$, $Sr=0.6$, $Du=0.2$, $r=0.01$, $kr=0.5$, $M=fw=0.1$.

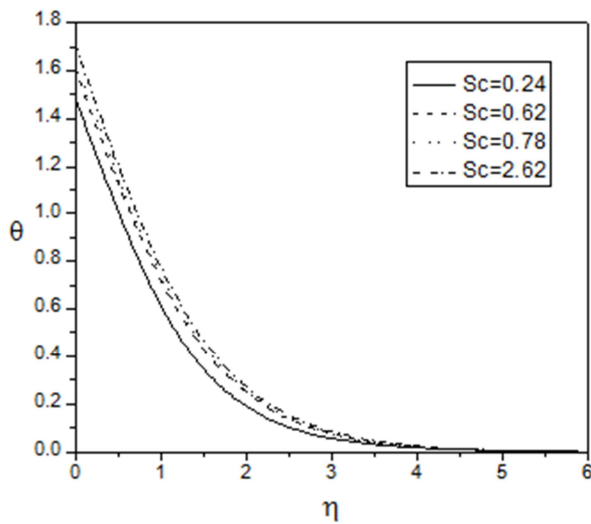


Fig. 15. Variation of the temperature θ with Sc for $Pr=0.71$, $Gr=Gc=1$, $Sr=0.6$, $Du=0.2$, $Q=r=0.01$, $kr=0.5$, $M=fw=0.1$.

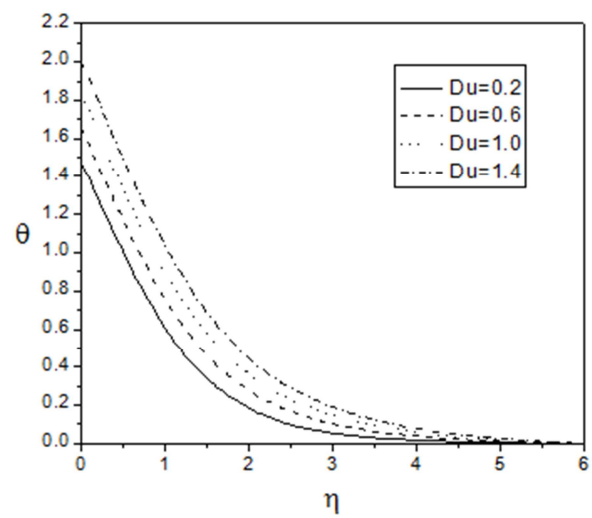


Fig. 18. Variation of the temperature θ with Du for $Pr=0.71$, $Sc=0.22$, $Gr=Gc=1$, $Sr=0.6$, $Q=r=0.01$, $kr=0.5$, $M=fw=0.1$.

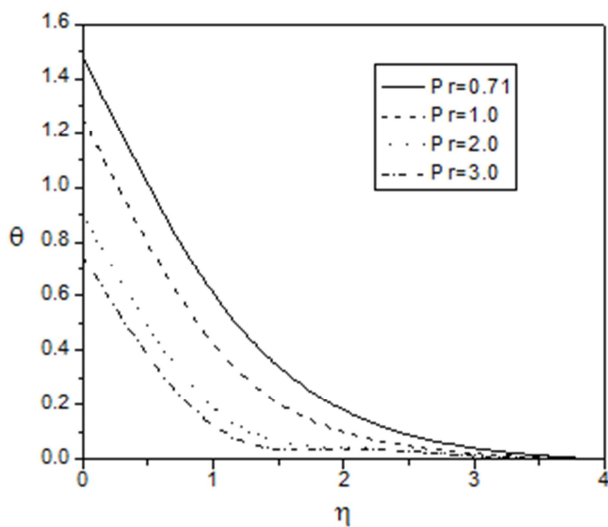


Fig. 16. Variation of the temperature θ with Pr for $Sc=0.22$, $Gr=Gc=1$, $Sr=0.6$, $Du=0.2$, $Q=r=0.01$, $kr=0.5$, $M=fw=0.1$.

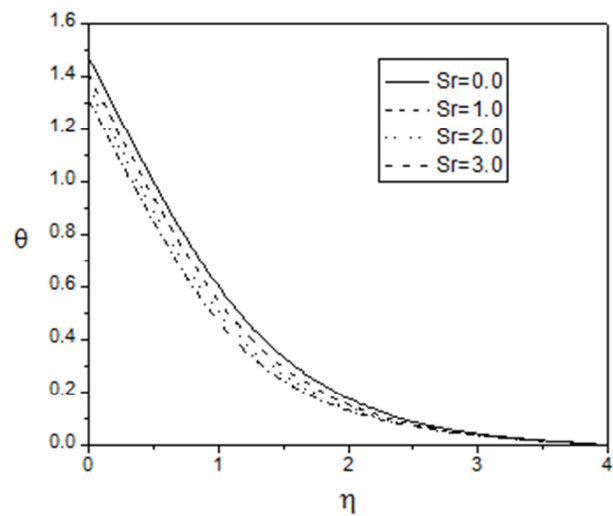


Fig. 19. Variation of the temperature θ with Sr for $Pr=0.71$, $Sc=0.22$, $Gr=Gc=1$, $Du=0.2$, $Q=r=0.01$, $kr=0.5$, $M=fw=0.1$.

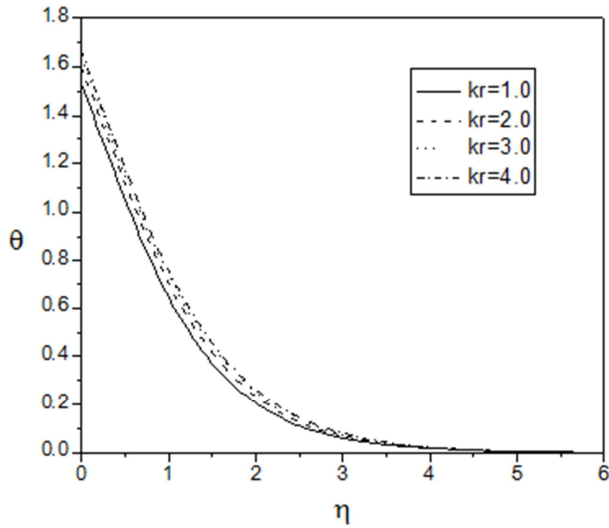


Fig. 20. Variation of the temperature θ with kr for $Pr=0.71$, $Sc=0.22$, $Gr=Gc=1$, $Sr=0.6$, $Du=0.2$, $Q=r=0.01$, $M=fw=0.1$.

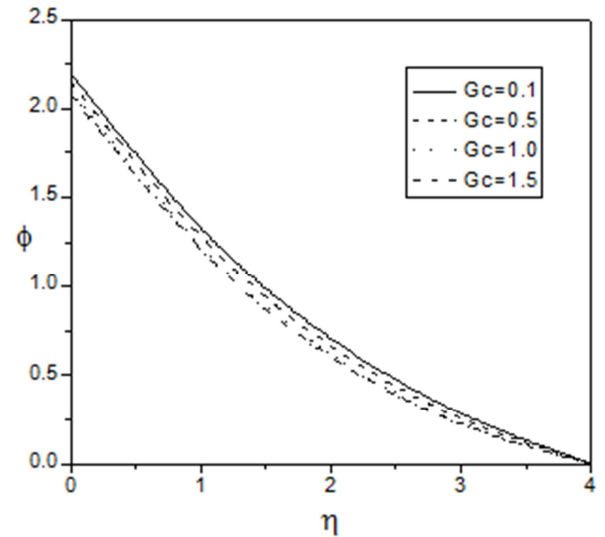


Fig. 23. Variation of the concentration ϕ with Gc for $Pr=0.71$, $Sc=0.22$, $Gr=1$, $Sr=0.6$, $Du=0.2$, $kr=0.5$, $Q=r=0.01$, $M=fw=0.1$.

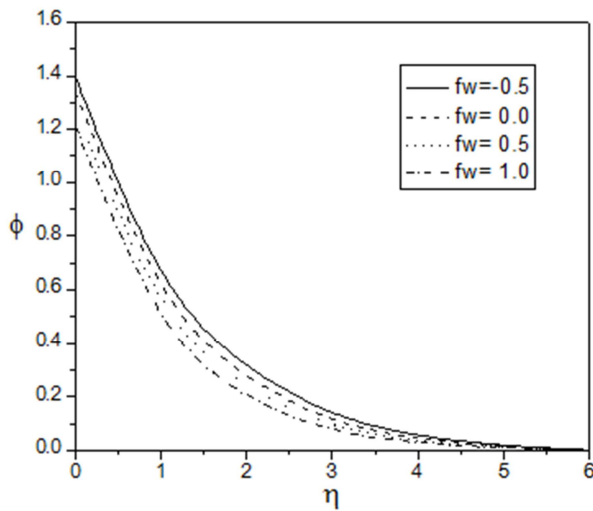


Fig. 21. Variation of the concentration ϕ with fw for $Pr=0.71$, $Sc=0.22$, $Gr=Gc=1$, $Sr=0.6$, $Du=0.2$, $kr=0.5$, $Q=r=0.01$, $M=0.1$.

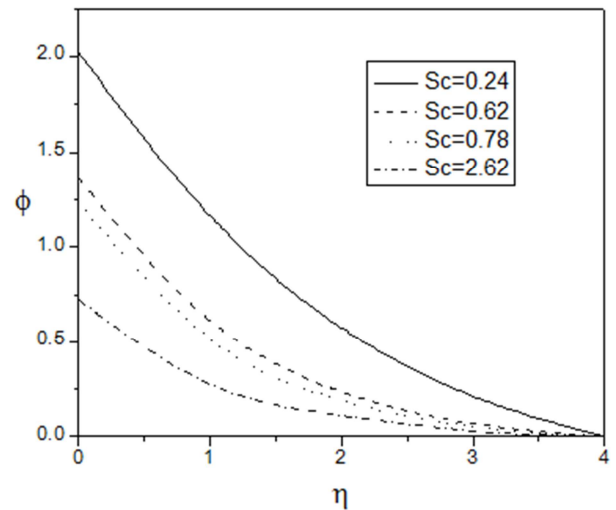


Fig. 24. Variation of the concentration ψ with Sc for $Pr=0.71$, $Gr=Gc=1$, $Sr=0.6$, $Du=0.2$, $kr=0.5$, $Q=r=0.01$, $M=fw=0.1$.

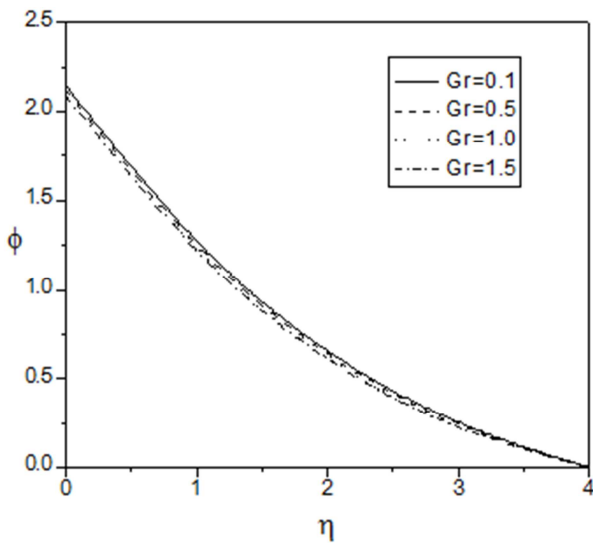


Fig. 22. Variation of the concentration ϕ with Gr for $Pr=0.71$, $Sc=0.22$, $Gr=1$, $Sr=0.6$, $Du=0.2$, $kr=0.5$, $Q=r=0.01$, $M=fw=0.1$.

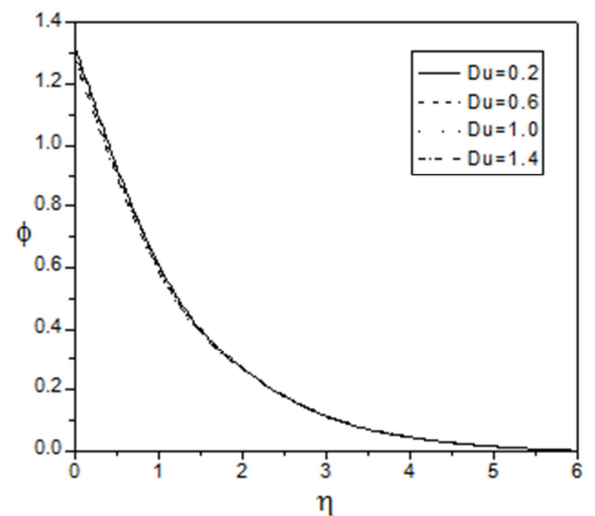


Fig. 25. Variation of the concentration ϕ with Du for $Pr=0.71$, $Sc=0.22$, $Gr=Gc=1$, $Sr=0.6$, $kr=0.5$, $Q=r=0.01$, $M=fw=0.1$.

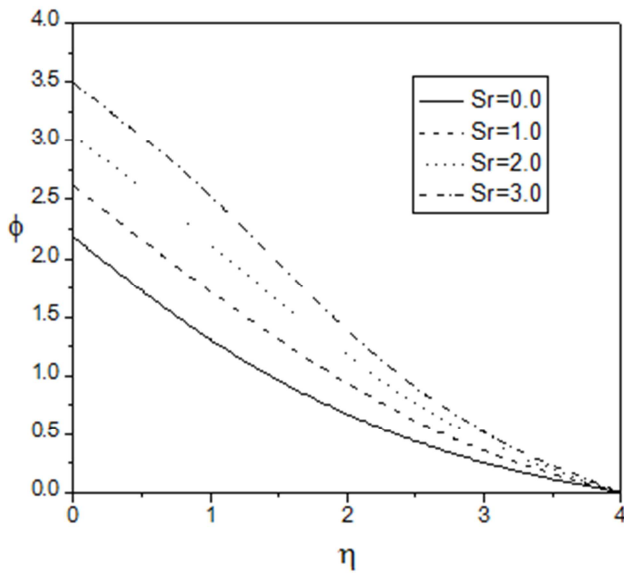


Fig. 26. Variation of the concentration ϕ with Sr for $Pr=0.71$, $Sc=0.22$, $Gr=Gc=1$, $Du=0.2$, $kr=0.5$, $Q=r=0.01$, $M=fw=0.1$.

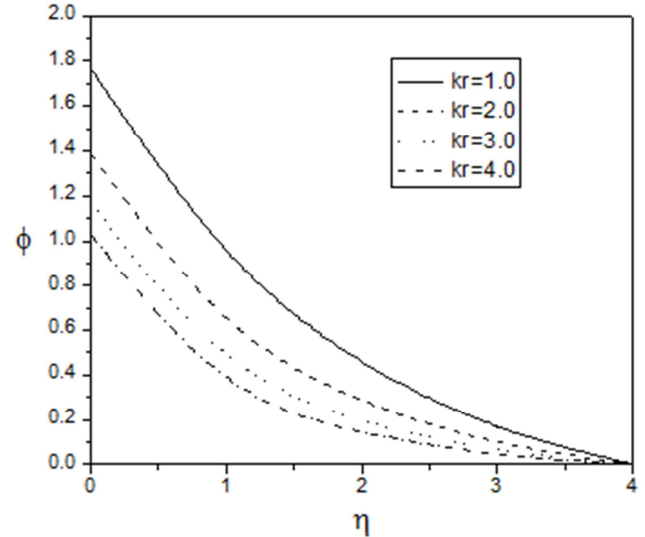


Fig. 27. Variation of the concentration ϕ with kr for $Pr=0.71$, $Sc=0.22$, $Gr=Gc=1$, $Sr=0.6$, $Du=0.2$, $Q=r=0.01$, $M=fw=0.1$.

Table 1. Variation of C_b , Nu and Sh at the plate with Gr , Gc , M , Sr , Du , Q , kr , fw for $r = 0.01$ and $Pr = 0.71$.

M	Gr	Gc	Sc	Sr	Du	Q	kr	Fw	C_f	Nu	Sh
0.01	1	1	0.22	0.6	0.2	0.01	0.5	0.1	0.425354	0.6231	0.753257
0.05	1	1	0.22	0.6	0.2	0.01	0.5	0.1	0.404656	0.621585	0.753019
0.1	1	1	0.22	0.6	0.2	0.01	0.5	0.1	0.379225	0.619725	0.752729
0.1	1.5	1	0.22	0.6	0.2	0.01	0.5	0.1	0.682416	0.6403	0.75601
0.1	2	1	0.22	0.6	0.2	0.01	0.5	0.1	0.96169	0.657636	0.758923
0.1	1	1.5	0.22	0.6	0.2	0.01	0.5	0.1	0.64016	0.639805	0.755903
0.1	1	2	0.22	0.6	0.2	0.01	0.5	0.1	0.896551	0.65815	0.758982
0.1	1	1	0.62	0.6	0.2	0.01	0.5	0.1	0.312685	0.609573	0.826341
0.1	1	1	0.78	0.6	0.2	0.01	0.5	0.1	0.293307	0.606906	0.852145
0.1	1	1	0.22	1.0	0.2	0.01	0.5	0.1	0.405195	0.62353	0.72905
0.1	1	1	0.22	2.0	0.2	0.01	0.5	0.1	0.470593	0.632679	0.675483
0.1	1	1	0.22	0.6	1.0	0.01	0.5	0.1	0.634505	0.479792	0.768973
0.1	1	1	0.22	0.6	1.4	0.01	0.5	0.1	0.752348	0.432614	0.777067
0.1	1	1	0.22	0.6	0.2	-1.0	0.5	0.1	0.0469438	1.0098	0.730834
0.1	1	1	0.22	0.6	0.2	0.1	0.5	0.1	0.440441	0.578361	0.756766
0.1	1	1	0.22	0.6	0.2	0.01	1.0	0.1	0.200716	0.595615	1.02619
0.1	1	1	0.22	0.6	0.2	0.01	1.5	0.1	0.119248	0.58494	1.24433
0.1	1	1	0.22	0.6	0.5	0.01	0.5	-0.5	0.815669	0.435261	0.714897
0.1	1	1	0.22	0.6	1.0	0.01	0.5	0.5	0.0710078	0.768309	0.782785

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